

Distinct Substrings

Time limit: 1.00 s

Memory limit: 512 MB

Count the number of distinct substrings that appear in a string.

Input

The only input line has a string of length n that consists of characters a–z.

Output

Print one integer: the number of substrings.

Constraints

- $1 \leq n \leq 10^5$

Example

Input	Output
abaa	8

Explanation: the substrings are a, b, aa, ab, ba, aba, baa and abaa.